

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 7 Summer 2025 (Supply.)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCOE703C - BIG DATA ANALYTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the main challenge in handling Big Data?
A. High storage cost B. Data consistency
C. Scalability and processing speed D. User interface design
- 2 Which of the following is a Big Data distribution package?
A. MATLAB B. Hadoop
C. Oracle D. Eclipse
- 3 In Spark, the main abstraction for data is called:
A. Hive B. RDD
C. Node D. Stream
- 4 The CAP Theorem does NOT include:
A. Consistency B. Accuracy
C. Availability D. Partition tolerance
- 5 Which of the following tools is widely used for Big Data streaming?
A. Hive B. Kafka
C. Pig D. Storm
- 6 What is a key requirement for Big Data streaming systems?
A. Batch processing B. Offline analysis
C. Low latency and high throughput D. SQL support
- 7 Mahout is a library for:
A. Data storage
B. Stream processing
C. Machine learning
D. Visualization
- 8 Which algorithm is used for classification in Mahout?
A. K-means B. Apriori C. Naive Bayes D. PageRank

- 9 GraphX is a component of:
 A. Cassandra
 B. Spark
 C. Mahout
 D. HDFS
- 10 MongoDB is a:
 A. Relational DBMS B. Document-oriented NoSQL database
 C. Column-store database D. Key-value store
- 11 In MongoDB, data is stored in:
 A. Tables B. Collections
 C. Files D. Buckets
- 12 Which command is used to insert a document in MongoDB shell?
 A. add() B. put()
 C. insert() D. update()
- Q2. Solve the following.
- A) Describe the Big Data Stack and its components. 6
- B) Explain enabling technologies that support Big Data. 6
- Q3. Solve the following.
- A) Explain the CAP Theorem and trade-offs in distributed systems. 6
- B) Discuss the differences between ACID and BASE models. 6
- Q4. Solve Any Two of the following.
- A) What are the challenges in real-time Big Data processing? 6
- B) Compare batch processing vs. stream processing. 6
- C) Describe Kafka and its role in real-time pipelines. 6
- Q5. Solve Any Two of the following.
- A) Compare Mahout and Spark MLlib for Big Data ML. 6
- B) Explain k-means clustering algorithm in Big Data context. 6
- C) What is GraphX? How does it support graph processing? 6
- Q6. Solve Any Two of the following.
- A) Explain collections and documents in MongoDB with examples. 6
- B) Describe the principles of schema design in MongoDB. 6
- C) Explain the key features of MongoDB. 6

*** End ***